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CELLULITIS

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CELLULITIS

Topic: Cellulitis — structured under Introduction, Objectives, Methodology and Review of Literature, with a conclusion and suggested references.

1. INTRODUCTION

Cellulitis is an acute bacterial infection involving the dermis and subcutaneous tissues. It typically presents with erythema, warmth, swelling and tenderness, often with poorly defined margins. It most commonly affects the lower limbs, although it can occur at other sites. Streptococci and Staphylococcus aureus are important causative organisms. Bacteria commonly enter through a break in the skin, including a wound, ulcer, insect bite, fissure or skin disease.

Cellulitis is primarily a clinical diagnosis. Early recognition and appropriate antimicrobial treatment are important to reduce progression and complications. Severe cases may be associated with systemic toxicity, bacteremia or sepsis. An abscess may coexist and require drainage, while suspected necrotizing soft-tissue infection requires urgent surgical assessment.

2. OBJECTIVES

General objective

To review the clinical features, diagnosis, management and complications of cellulitis relevant to undergraduate medical education and General Surgery.

Specific objectives

• To define cellulitis and describe its etiology and predisposing factors.

• To explain the pathogenesis and clinical presentation of cellulitis.

• To outline investigations used when clinically indicated.

• To review principles of medical and surgical management.

• To differentiate cellulitis from abscess, erysipelas and necrotizing soft-tissue infection.

• To identify important complications, warning signs and preventive measures.

3. METHODOLOGY

This document is a narrative literature review prepared for MBBS-level academic study. It synthesizes standard clinical concepts commonly taught in General Surgery and infectious disease education. It is not a primary research study, systematic review, clinical trial or patient-data analysis.

Approach

• **Topic scope:** definition, causative organisms, risk factors, pathogenesis, clinical features, diagnosis, treatment, complications and prevention.

• **Information synthesis:** established clinical principles are organized into exam-oriented sections and comparison tables.

• **Clinical framing:** uncomplicated cellulitis is distinguished from abscess and potentially life-threatening deep soft-tissue infection.

• **Limitations:** no formal database search, predefined inclusion/exclusion criteria, statistical analysis or independent quality appraisal was performed for this educational summary.

Important: For formal research submission, replace or expand this section with the databases searched, search terms, date range, selection criteria and appraisal method actually used. Do not claim a search was performed unless it was.

4. REVIEW OF LITERATURE

4.1 Definition and classification

Cellulitis is a diffuse infection of the dermis and subcutaneous tissue. It is commonly described as non-purulent when no drainable collection is present, or purulent when pus or an associated abscess is identified. Erysipelas is a more superficial infection with a typically raised, sharply demarcated margin.

4.2 Etiology and risk factors

• **Common organisms:** *Streptococcus pyogenes* and *Staphylococcus aureus*.

• **Exposure-related organisms:** *Pasteurella* species after animal bites and *Eikenella corrodens* after human bites; other organisms may be relevant after water exposure.

• **Predisposing factors:** breaks in skin, ulcers, fungal infection between toes, eczema, diabetes mellitus, edema, venous insufficiency, lymphedema and immunosuppression.

• **Recurrence:** persistent edema, skin disease and untreated portals of entry can contribute to repeated episodes.

4.3 Pathogenesis

Bacteria enter through a disrupted skin barrier and multiply within the dermis and subcutaneous tissue. The host inflammatory response causes vasodilation, increased capillary permeability and migration of inflammatory cells. These changes produce erythema, warmth, edema and tenderness. If infection progresses, lymphangitis, lymphadenitis, abscess, bacteremia or sepsis may occur.

4.4 Clinical features

• Local erythema, warmth, swelling and tenderness.

• Usually poorly defined borders in cellulitis.

• Possible fever, chills, malaise and tachycardia in more severe disease.

• A portal of entry may be visible, such as a wound, ulcer, fissure or bite.

• Red streaking may suggest lymphangitis; regional lymph nodes may be tender.

4.5 Diagnosis and investigations

Diagnosis is usually clinical. Assessment should establish the extent and speed of spread, systemic stability, likely portal of entry and whether an abscess or deeper infection is present. CBC, inflammatory markers, glucose, renal function and other tests may be useful in selected patients according to severity and comorbidities. Blood cultures are not routinely required in every uncomplicated case but may be considered in severe systemic illness, immunocompromise or unusual exposures. Ultrasound can help identify an occult abscess; advanced imaging is reserved for selected deep or complex infections.

4.6 Differential diagnosis

Condition	Key distinguishing features
Cellulitis	Diffuse erythema, warmth, edema and tenderness; poorly defined margins.
Abscess	Localized collection of pus; may be fluctuant and may require incision and drainage.
Erysipelas	More superficial infection, typically with a raised, sharply demarcated edge.
Necrotizing soft-tissue infection	Rapid progression, severe pain out of proportion, systemic toxicity, bullae or tissue necrosis.
Deep vein thrombosis	Pain and limb swelling that may mimic cellulitis; evaluate when clinically suspected.

4.7 Management

Treatment depends on severity, likely organisms, allergy history, local antimicrobial resistance and patient-specific factors. Core principles include appropriate antibiotics, elevation of the affected limb, pain control, hydration when needed and treatment of the portal of entry. Mild uncomplicated cases may be treated with oral antibiotics; intravenous therapy and hospital care may be required for severe illness, systemic toxicity, significant comorbidity or inability to take oral medication.

An associated abscess requires assessment for drainage. Uncomplicated cellulitis without a collection is not routinely treated by incision and drainage. Suspected necrotizing infection is a surgical emergency requiring urgent specialist assessment, resuscitation, broad-spectrum intravenous antibiotics and prompt operative management when indicated. Clinical deterioration or failure to begin improving should prompt reassessment of the diagnosis, antimicrobial choice and possibility of an abscess or deeper infection.

4.8 Complications

• Abscess formation.

• Lymphangitis and regional lymphadenitis.

• Bacteremia and sepsis.

• Rarely, spread to deeper tissues, including necrotizing infection, osteomyelitis or septic arthritis.

• Recurrent cellulitis and persistent edema or lymphatic dysfunction.

4.9 Prevention and prognosis

Prevention includes appropriate wound care, treatment of fungal infection and chronic skin disease, good diabetic foot care, and management of edema and venous insufficiency. Most uncomplicated cases improve with appropriate treatment, although residual swelling or discoloration may persist after the acute infection settles. Rapid progression, disproportionate pain, bullae, dark skin changes, confusion or faintness warrant urgent medical assessment.

5. CONCLUSION

Cellulitis is a common bacterial infection of the dermis and subcutaneous tissue. Recognition of its characteristic clinical features, assessment of severity and selection of appropriate antimicrobial therapy are central to management. Identifying portals of entry and predisposing conditions helps prevent recurrence. Clinicians must assess for abscess formation and recognize warning signs of necrotizing infection or sepsis, which require urgent intervention.

6. References

1. Williams NS, O'Connell PR, McCaskie AW. *Bailey & Love's Short Practice of Surgery*. 27th ed.

2. Townsend CM, Beauchamp RD, Evers BM, Mattox KL. *Sabiston Textbook of Surgery*. 21st ed.

3. Bhat Sriram M. *SRB's Manual of Surgery*. 6th ed.

4. Brunicardi FC, Andersen DK, Billiar TR, Dunn DL. *Schwartz's Principles of Surgery*. 11th ed.

5. Zinner MJ, Ashley SW, McFadden DW. *Maingot's Abdominal Operations*. 13th ed.

Educational summary only. For patient care, follow current local clinical protocols and individual medical assessment.